S/N	Arduino Mega Pin / Interface	I/O Type	Code Variable	Device	Signal / Function	Configuration / Notes
1	A0	Analog input	ir_sensor_1	QRE1113 #1	IR line sensor output	analogRead(A0)
2	A2	Analog input	ir_sensor_2	QRE1113 #2	IR line sensor output	analogRead(A2)
3	A4	Analog input	ir_sensor_3	QRE1113 #3	IR line sensor output	analogRead(A4)
4	A6	Analog input	ir_diode_transistor_1	OPB706B #1	IR phototransistor output	analogRead(A6)
5	A8	Analog input	ir_diode_transistor_2	OPB706B #2	IR phototransistor output	analogRead(A8)
6	A10	Analog input	ir_diode_transistor_3	OPB706B #3	IR phototransistor output	analogRead(A10)
7	A13	Analog input	ir_distance_sensor	Sharp GP2Y0A02YK	IR distance sensor output	ADC converted to distance by sharpDistance()
8	D7	Digital output	servo1_pitch	SG90 #1	Pitch servo control	Servo.attach(7)
9	D8	Digital output	servo2_yaw	SG90 #2	Yaw servo control	Servo.attach(8)
10	D26	Digital output	right_motor_fwd	H-bridge / right motor	Right motor direction	HIGH = forward, LOW = reverse
11	D4	PWM output	right_motor_speed	H-bridge / right motor	Right motor speed	analogWrite(); +50 software compensation
12	D24	Digital output	left_motor_fwd	H-bridge / left motor	Left motor direction	HIGH = forward, LOW = reverse
13	D5	PWM output	left_motor_speed	H-bridge / left motor	Left motor speed	analogWrite()
14	D2	Digital input / interrupt	right_wheel_encoder	SEN0038 right encoder	Right wheel encoder pulse	INPUT_PULLUP; RISING interrupt
15	D3	Digital input / interrupt	left_wheel_encoder	SEN0038 left encoder	Left wheel encoder pulse	INPUT_PULLUP; RISING interrupt
16	D20 (SDA)	I2C	Wire / IMU	IMU subsystem	I2C data	Wire.begin(); Mega hardware SDA
17	D21 (SCL)	I2C	Wire / IMU	IMU subsystem	I2C clock	Wire.begin(); Mega hardware SCL
18	USB / Serial0	Serial	Serial	Raspberry Pi / host	Command and telemetry interface	9600 baud